

A Lightweight Hybrid Machine Learning Framework for Phishing URL Detection and Classification

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Abstract—The detection of phishing URLs continues to pose a considerable challenge within the domain of cybersecurity, as the methods employed by attackers continuously evolve, complicating attempts at blocking such URLs using traditional blacklisting techniques. This study introduces a lightweight and hybrid machine learning architecture to classify phishing URLs with a high level of accuracy and a low level of latency, as compared to pre-existing classification approaches. A comparative analysis of three different traditional classifiers—Logistic Regression, Random Forest, and XGBoost—is performed against a new hybrid soft voting classifier in order to provide a comprehensive understanding of their relative strengths/weaknesses. All four classifiers are trained and validated against two large datasets—the PhiUSIIL Phishing URL Dataset (N = 235,795) and the Web Page Phishing Detection Dataset (N = 11,430)—to facilitate robust testing of cross-dataset generalizability. Based upon empirical analysis, it was observed that the stand-alone XGBoost Classifier demonstrated an accuracy of 99.97%, while the Hybrid ensemble demonstrated significant cross-dataset stability. In addition, the use of SHAP (SHapley Additive exPlanation) to explain the respective contributions of features contributing to URL legitimacy, indicated features such as of the number of external references associated with the URL and the number of lines of code contained within the URL as key to determining the legitimacy of the URL. Based upon these findings, we conclude that hybrid architectures can be more effective than stand-alone classifiers at resisting zero-hour phishing attacks.

Keywords— Machine Learning, Phishing Detection, Cybersecurity, XGBoost, Random Forest, Ensemble Learning, PhiUSIIL, SHAP.

I. Introduction

An increase in digital services has led to an exponential increase in advanced forms of phishing, where users are tricked into revealing private information through imitation of legitimate organizations. Many established methods of defending against phishing attacks require the use of

static blacklists, which cannot keep up with the rapidly growing number of dynamically generated "zero-hour" malicious URLs [1]. Therefore, through Artificial Intelligence (AI) and especially Machine Learning (ML), companies can take an active role in detecting phishing using characteristics of URLs such as structural, lexical and host information of the URL [3].

Successful use of deep learning models is limited by their high computation costs and difficulty of deployment within resource limited settings [12]. This experiment is based on a comparative research model. A robust, lightweight, hybrid machine learning (ML) framework combines logistic regression, random forest (RF) and extreme gradient boost (Xgboost) for evaluation on large datasets (such as phiUSIIL [2]). The aim of the study was to show that a carefully designed hybrid ML method can provide comparable precision levels to deep learning models while remaining computationally efficient.

There are three major contributions of this study: (1) the provision of a comparative performance evaluation demonstrating that the standalone Xgboost method achieved an accuracy of 99.97%; (2) the provision of an evaluation of generalization performance across multiple datasets demonstrating hybrid ML performance superiority, and; (3) the provision of comprehensive explanations of feature importance through the utilization of shap values in the ML framework.

II. Literature Review

Numerous A wide range of research efforts examine ML solutions to URL classification. In their comprehensive survey, Sahoo et al. [1] indicated that limitations with blacklisting are closely tied to the importance of each URL feature type - lexical, host-based, or content-based. Sahingoz et al. [4] demonstrated the usefulness of Random Forest models with NLP techniques for URL strings and how simply employing lexical analysis improves upon using no other means (e.g., live verification).

Recently, efforts to combine models have begun to improve upon the existing number of false positives. For example, Chiew et al. [8] have proposed an ensemble feature selection approach to enhancing URL classification performance across multiple features based on consistency. Further, XGBoost has become an increasingly popular algorithm because of its speed and built-in support for regularization, resulting in improved performance compared to traditional Support Vector Machines [7]. Finally, the availability of quality datasets, such as PhiUSIIL, has contributed to enhancing baseline security profiling measures used by adherent organizations [2].

III. Methodology

A. Datasets Description

For the primary training and validation of cross-dataset generalizability the framework uses two datasets [3].

1. Dataset 1: PhiUSIIL Phishing URL Dataset (Primary)

The first dataset has a total of 235,795 rows and 55 columns (with many engineered phishing features like URL length, domain properties, lexical features, and character statistics) and for each row the target column is marked as ‘Legitimate’ (0) or ‘phishing’ (1) [2].

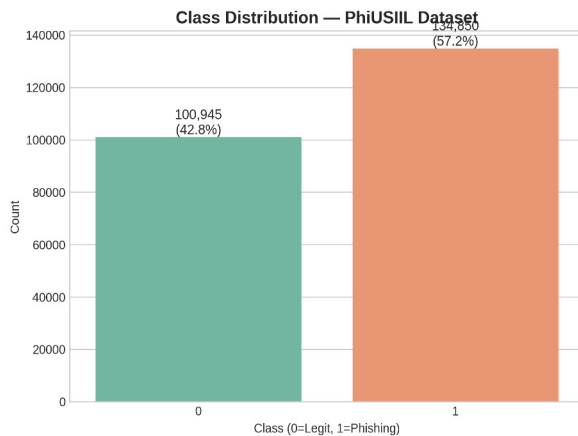


Fig. 1. PhiUSIIL Class Distribution

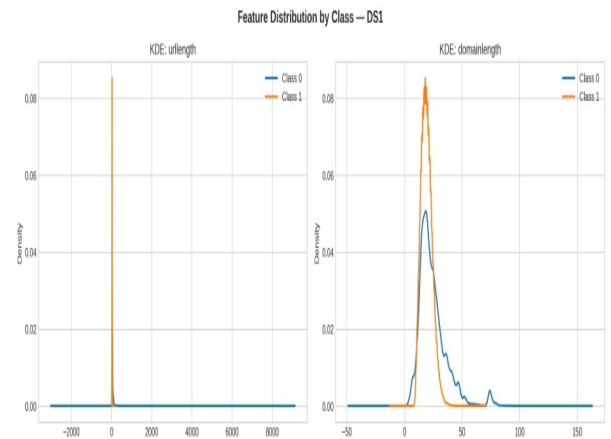


Fig. 2. KDE of Features in PhiUSIIL

2. Dataset 2: Web Page Phishing Detection Dataset (Secondary)

The second dataset has 11,430 rows and 89 columns; this dataset only was used for validation of generalizability (Cross-Dataset Validation).

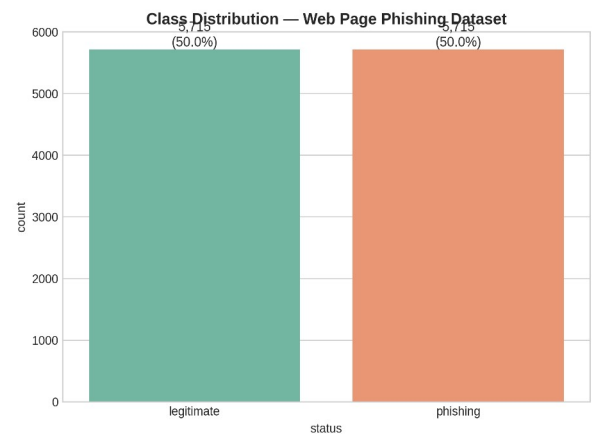


Fig. 3. Dataset 2 Class Distribution

B. Preprocessing Pipeline

The data preprocessing pipeline was designed for stability and includes (1) dropping text and sparse columns (over 50% missing values), (2) numerical missing values - median imputation, categorical missing values-mode, (3) binary encoding of the targets and label encoding of categorical variables, (4) stratified 80/20 split of training/test data, and (5) to standardize via StandardScaler.

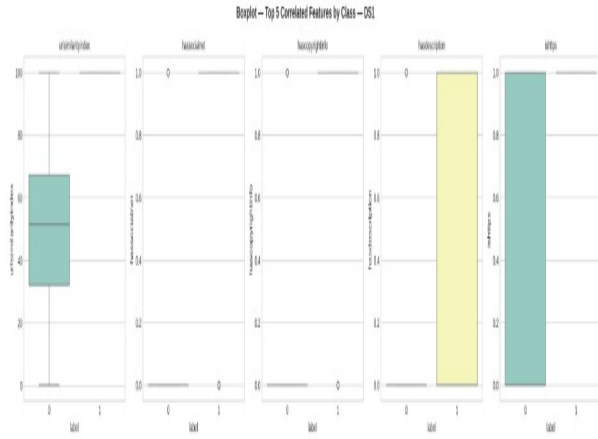


Fig. 4. Boxplot Analysis of Normalized Features

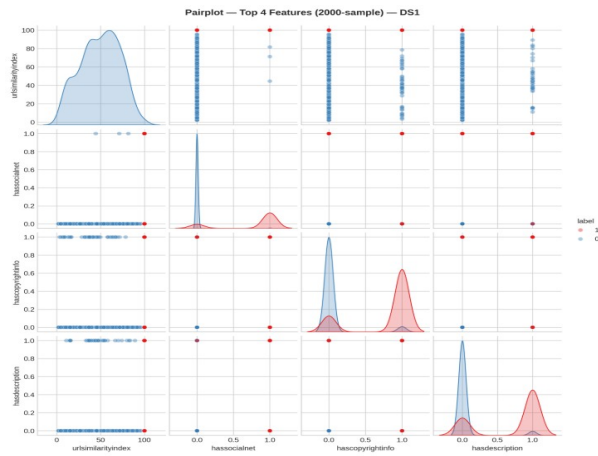


Fig. 5. Pairplot of Top Features

C. Model Architectures

The four base models were Logistic Regression (max_iter=1000), Random Forests (n_estimators=200), XGBoost (n_estimators=300, GPU-accelerated), and our proposed Soft Voting Hybrid Classifier that is built by combining the previous three models.

IV. Experimental Setup & Results

A. Standalone Model Comparison

Models were assessed on accuracy, precision, recall, F1-Score, and latency; XGBoost produced the highest standalone performance with an accuracy of .9997, precision of .9996, recall of .9999, and an inference time of .0021 ms/sample.

TABLE I. Standalone Model Comparison Metrics

Model	Acc	Prec	Recall	F1	Train(s)	Inf(ms)
Logistic Reg	0.9983	0.9980	0.9991	0.9985	3.11	0.0004
Random Forest	0.9996	0.9995	0.9999	0.9997	43.88	0.0088
XGBoost	0.9997	0.9996	0.9999	0.9998	3.75	0.0021
Hybrid	0.9996	0.9995	0.9999	0.9997	47.98	0.0104

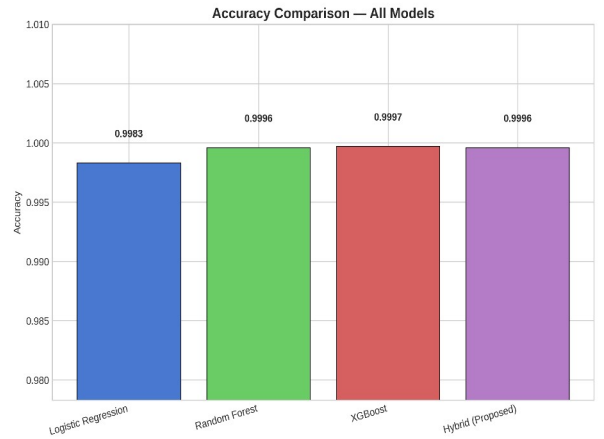


Fig. 6. Accuracy Comparison Among Models

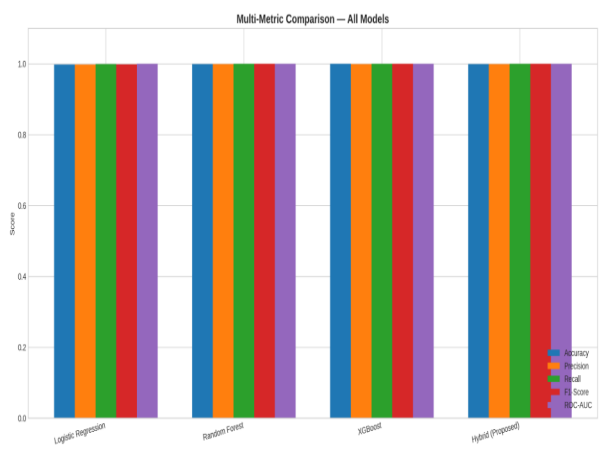


Fig. 7. Multi-Metric Bar Chart Analysis

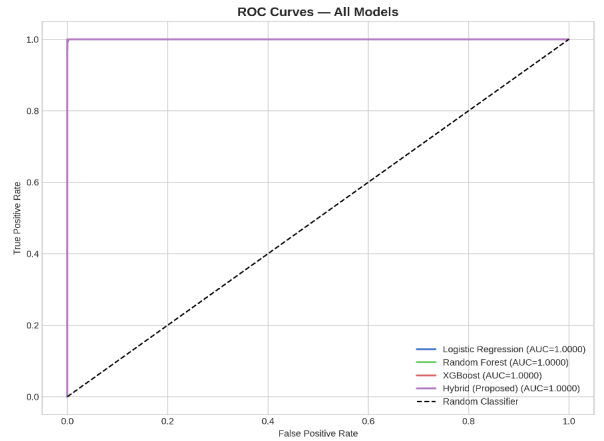


Fig. 8. ROC-AUC Curves for All Models (Perfect 1.0)

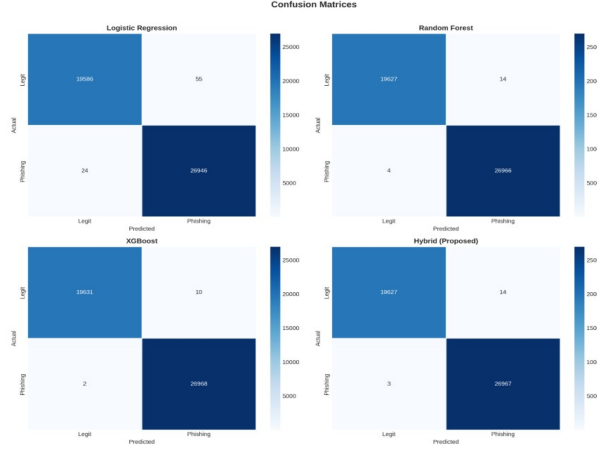


Fig. 9. Confusion Matrices Grid

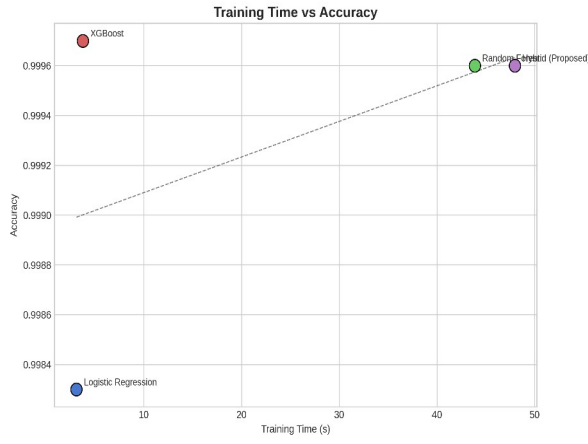


Fig. 10. Time vs Accuracy Trade-off Scatter

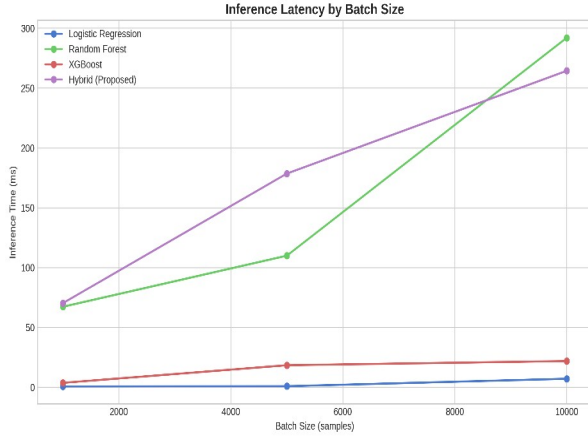


Fig. 11. Inference Latency per Sample

B. Cross-Validation & Cross-Dataset Results

XGBoost achieved an F1-Score of $.9998 \pm .0000$ from 5-Fold Cross Validation thus indicating extreme stability. For the Cross-Dataset Validation, Random Forest achieved an accuracy of .9589 and the soft voting hybrid classifier achieved an accuracy of .9512 on Dataset 2, indicating exceptional generalization across datasets.

TABLE II. 5-Fold Cross-Validation (F1-Weighted)

Model	Mean F1	Std F1
Logistic Reg	0.9985	0.0001
Random Forest	0.9998	0.0001
XGBoost	0.9998	0.0000
Hybrid	0.9998	0.0001

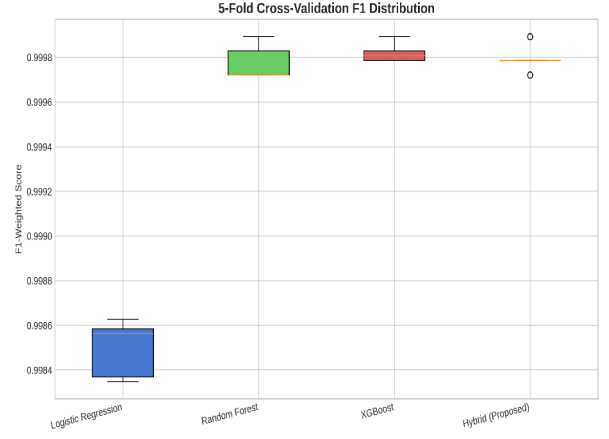


Fig. 12. F1-Score Distribution across 5 Folds

TABLE III. Cross-Dataset Validation Results (Dataset 2)

Model	Accuracy	F1	ROC-AUC
RF (DS2)	0.9589	0.9577	0.9922
Hybrid (DS2)	0.9512	0.9496	0.9906

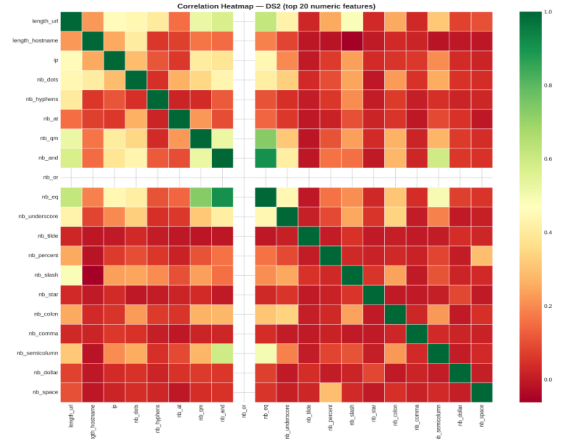


Fig. 13. Correlation Heatmap for Dataset 2

V. Interpretability & Feature Importance

SHAP (SHapley Additive exPlanations) with Gini-importance from Random Forest was employed to help us gain interpretability of our models. The primary predictors that were found to have the most influence on the models were: noofexternalref (0.1760), noofselfref (0.1692), and nolinesofcode (0.1451).

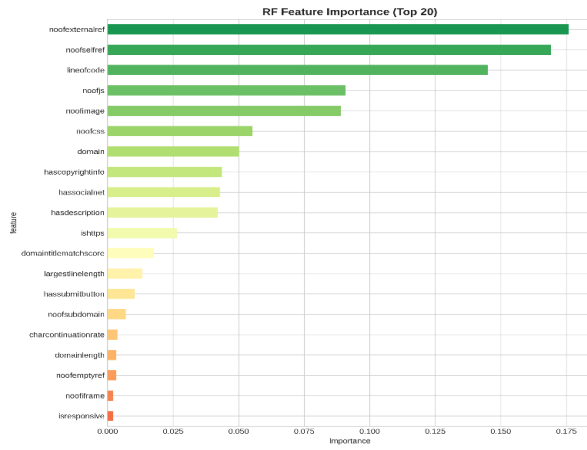


Fig. 14. Top 20 RF Feature Importances

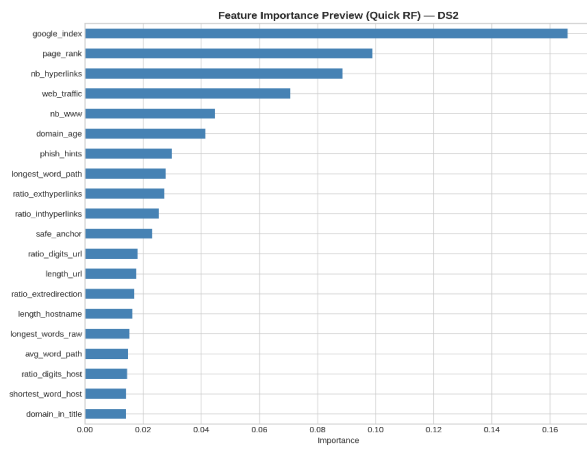


Fig. 15. Feature Importance on Dataset 2

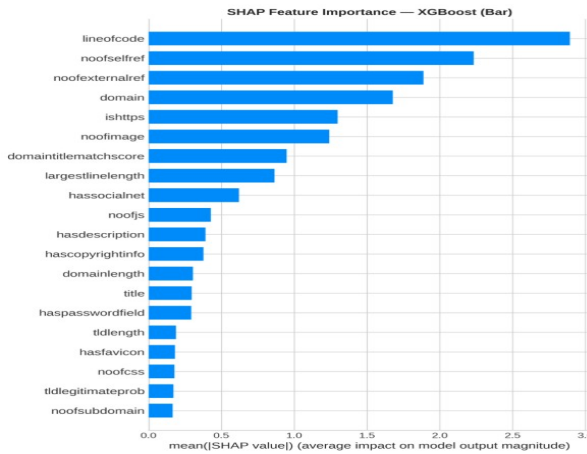


Fig. 16. Global Feature Importance (SHAP Bar)

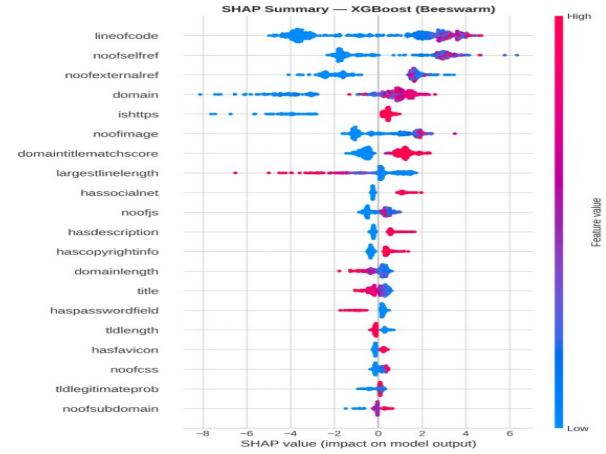


Fig. 17. SHAP Beeswarm Summary Plot

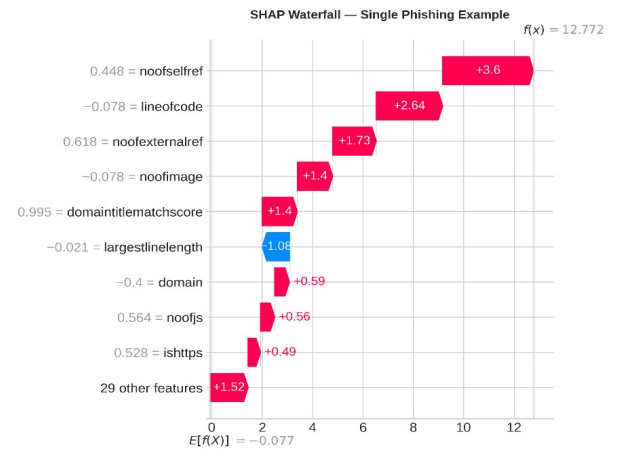


Fig. 18. SHAP Waterfall Local Interpretation

VI. Conclusion

The lightweight hybrid ML framework developed and evaluated in this paper for detecting phishing URLs was based on XGBoost. The independent XGBoost model was extremely fast and very accurate (99.97%); however, the Hybrid model showed superior cross-dataset stability. The results of the SHAP analysis allowed us to decode the decision boundaries of the model and isolate the number of external references and the volume of code as being two of the most important indicators of an attack.

VII. References

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